

INFORMATION VISUALISATION

COMPSCI5099 · Exam Cheatsheet · University of Glasgow

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DATA ATTRIBUTE TYPES

Type	Operations	Example
Nominal	= ≠ only	Country, Blood type, Ward
Ordinal	= ≠ < >	Low/Med/High, Grades
Quantitative	= ≠ < > + −	Age, Weight, Salary

SHNEIDERMAN'S 7 DATA TYPES — 1-2-3-T-M-Tr-N

Type	Example	Good Vis
1D Linear	Text, lists	Timeline
2D Planar	Maps, plans	Map / scatter
3D Volumetric	MRI, molecules	3D render
Temporal	Stock prices	Line chart
Multi-dim	DB records	Parallel coords
Tree	File system	Treemap
Network	Social graph	Node-link

BERTIN'S VISUAL VARIABLES

Variable	Nominal	Ordinal	Quant	Select?
Position	✓	✓	✓ BEST	✓
Size	■	✓	✓	✓
Shape	✓	✗	✗	✗ ONLY ONE
Hue (colour)	✓	✗	✗	✓
Value (lightness)	✗	✓	■	✓
Orientation	■	✗	✗	✓
Texture	✓	■	✗	✓

★ **Shape = ONLY non-selective.** Position & Size = ONLY quantitative variables.
Hue → nominal. Value → ordinal. Position → quantitative.

VISUALISATION PIPELINE

Stage	What happens
1. Raw Data	Original source
2. Data Transform	Clean, filter, aggregate
3. Visual Mapping ★	Map attributes → marks & channels (MOST CRITICAL)
4. View Transform	Zoom, pan, navigate
5. Rendering	Draw pixels on screen

SHNEIDERMAN'S MANTRA + 7 TASKS

"Overview first, zoom and filter, then details-on-demand" — Shneiderman (1996)

Task	When to use
① Overview	Always first — show all data
② Zoom	User sees interesting area
③ Filter	Remove irrelevant items
④ Details-on-demand	Hover/click for full record
⑤ Relate	Show connections between items
⑥ History	Undo / replay exploration
⑦ Extract	Export filtered subset

STEVENS' LAW + ACCURACY RANKING

$P = k \times S^n$ — Perceived magnitude follows a power law

Rank	Channel	n	Why / Chart
① BEST	Position on common scale	≈1.0	Bar chart, Scatter
②	Position, unaligned	≈1.0	Grouped charts
③	Length / Height	≈1.0	Bar (the bar itself)
④	Angle / Slope	—	Pie chart ← WHY PIES ARE BAD
⑤	Area	≈0.7	Bubble (underestimated ~30%)
⑥	Volume	≈0.6	3D charts ← avoid
⑦ WO RST	Colour saturation	≈0.5	Heatmap

COLOUR THEORY

Map Type	Use for	Example
Sequential	Ordered, one direction	Light→dark blue for rainfall
Diverging	Meaningful zero/midpoint	Blue–white–red for temp anomaly
Categorical	Nominal (≤8 categories)	Countries, product types

Rainbow map = BAD. Not perceptually uniform → creates false boundaries. Use: Viridis

Colour Rule

Never red+green as only differentiator (8% males colour-blind)
Use redundant encoding: colour + shape or colour + texture
Limit categorical hues to ≤ 8 colours
CIELAB is the only perceptually uniform colour space

PRE-ATTENTIVE PROCESSING			CHART TYPES — WHEN TO USE										
Features detected in <200ms without conscious attention — the "pop-out" effect. <table><tr><th>Pre-attentive ✓ (pop-out)</th><th>NOT pre-attentive ✗</th></tr><tr><td>Colour (single feature)</td><td>Colour + shape together</td></tr><tr><td>Size / orientation</td><td>Any conjunction of 2 features</td></tr><tr><td>Motion, enclosure</td><td>Requires serial scanning</td></tr></table>			Pre-attentive ✓ (pop-out)	NOT pre-attentive ✗	Colour (single feature)	Colour + shape together	Size / orientation	Any conjunction of 2 features	Motion, enclosure	Requires serial scanning	Chart	Use for	Avoid when
			Pre-attentive ✓ (pop-out)	NOT pre-attentive ✗									
			Colour (single feature)	Colour + shape together									
			Size / orientation	Any conjunction of 2 features									
			Motion, enclosure	Requires serial scanning									
Bar	Categorical comparison	Trend over time											
Line	Trends over time	Categorical x-axis											
Scatter	Correlation, clusters	Categorical data											
Histogram	Distribution shape	Comparing groups											
Gestalt Principles:			Box plot	Distribution summary	Bimodal data (hides it)								
			Pie	Rough proportions (≤5)	Any precise comparison								
Principle	Means	Use	Heatmap	Large matrix patterns	Precise values needed								
Proximity	Close = grouped	Small multiples	Parallel Coords	Multidimensional data	Many items (clutter)								
Similarity	Alike = grouped	Colour coding											
Continuity	Eye follows paths	Line charts											
Enclosure	Inside = grouped	Panel borders											
Connectedness	Linked = grouped	Network edges											

MULTIDIMENSIONAL TECHNIQUES				NETWORK VISUALISATION		
Technique	Dims	Strength	Weakness		Node-Link	Matrix
Parallel Coords	10–20	All dims at once. Correlations.	Adjacent axes only. Clutter.	Best for	Sparse graphs. Path tracing.	Dense graphs. Connection check.
SPLOM	8–12	All pairwise scatter plots.	n^2 plots. Small cells.	Strength	Clusters visible. Intuitive.	No edge crossings.
Star/Radar	5–10	Compare individual profiles.	Area distorted by ordering.	Weakness	Hairball with dense graphs.	Hard to trace paths.
Heatmap	50+	Compact. Large matrices.	Colour limits precision.	"Are A & B connected?"	Hard — visual search	Easy — find row A, col B
t-SNE / UMAP	100+	Overview of very high-D data.	Loses dim interpretability.	Find communities	Clusters emerge visually	Dense diagonal blocks
TREE VISUALISATION						

Parallel coords: parallel lines = positive correlation. X-shape = negative. SPLOM formula: n dims → $n \times (n-1)$ plots. 9 dims = 72 plots.				Technique	Best for
				Node-link tree	Small hierarchies. Clear structure.
				Treemap	Large hierarchy + quantitative attribute (file sizes)
				Sunburst	Hierarchical navigation
				Icicle plot	Wide hierarchies with clear levels
Dim Reduction	Preserves	Best for			
PCA	Global variance	Linear structure			
t-SNE	Local clusters only	Cluster vis			
UMAP	Local + global	Most tasks			
MDS/Spring	Pairwise distances	Similarity-based layout			

TEMPORAL DATA VISUALISATION		LARKIN & SIMON (1987) — Diagrams vs Text	
Technique	Use when	Concept	Definition
Line chart	Single time series, trends	Informational Equivalence	Same information can be derived from both
Stacked area	Composition over time	Computational Equivalence	Same cognitive effort needed to use both
Small multiples	Comparing patterns across categories		
Calendar heatmap	Cyclic patterns (day-of-week effects)	3 Advantages of Diagrams:	
Gantt chart	Intervals, durations, scheduling	Advantage	Means
Sparklines	Tiny trends embedded in tables	Locality	Related info grouped spatially → less search
Animation vs Small Multiples: Animation suffers from change blindness — comparison. Always use small multiples for analysis.		Minimise labelling	Location itself carries meaning
		Perceptual inference	Visual system does the work (e.g. see lines cross)
Key exam point: Representations can be informationally equivalent but NOT computationally — not adding information.			

HEER — GENERALISED SELECTION		
	Dynamic Queries	Query Relaxation
Direction	Broad → narrow	Specific → broad
Method	Sliders / toggles	Click item, generalise
Approach	Top-down	Bottom-up

EVALUATION — Munzner's Nested Model			TUFTE'S RULES + COMMON PITFALLS	
Level	Question	Threat if wrong	Rule	Meaning
1. Domain	What do users need?	Solving wrong problem	Data-ink ratio	Maximise. Remove all non-data ink.
2. Abstraction	Right data types & tasks?	Wrong model of data	Chartjunk	Delete: 3D, heavy grids, decorations
			Lie factor	= graphic effect ÷ data effect. Should be ≈1.0
3. Encoding	Right marks & channels?	Ineffective visualisation	Small multiples	Better than animation for comparison
			No unjustified 3D	3D = occlusion + distortion. Avoid.
4. Algorithm	Efficient implementation?	Too slow to use	Eyes beat memory	Put things side-by-side, not sequential
			✗ Pitfall	✓ Fix
Eval Method	Use when...		Truncated y-axis	Always start bar charts at 0
Controlled experiment	Prove A is faster/better than B (statistics)		Pie chart for comparison	Use bar chart (position > angle)
Think-aloud / interview	Understand WHY users struggle		Rainbow colour map	Use Viridis / Inferno / Magma
Heuristic evaluation	Quick expert design review		Hue for ordinal data	Use colour value (lightness)
Case study	Real users, real tasks, ecological validity		Size for nominal data	Use hue or shape instead
			Dual y-axes	Use two separate charts

KEY PEOPLE — Who said what

Person	Year	One-line contribution	Magic phrase / exam hook
Bertin	1967	Visual variables: position + 6 retinal. Assoc/selective/ordered/quant properties.	"Shape is the only non-selective variable"
Shneiderman	1996	"The Eyes Have It" — mantra + 7 tasks × 7 data types.	"Overview first, zoom and filter, details-on-demand"
Tufte	1983+	Data-ink ratio, chartjunk, lie factor, small multiples.	"Above all else, show the data"
Munzner	2015	Nested model (4 levels). Marks & channels. Effectiveness ranking.	"Expressiveness + Effectiveness"
Cleveland & McGill	1984	Empirical perceptual accuracy hierarchy: position > length > angle > area.	"Position on common scale is most accurate"
Larkin & Simon	1987	"Why a Diagram is (Sometimes) Worth Ten Thousand Words." Info vs computational equiv.	"Same info, different cognitive effort"
Heer et al.	~2008	Generalised selection via query relaxation. Selection = declarative query.	"Bottom-up, example-driven query building"
Treisman	1985	Feature Integration Theory. Pre-attentive = parallel. Conjunction = serial.	"Single features pop out; combinations do not"
Stevens	1957	Power Law $P=kS^n$. Area ($n=0.7$) underestimated. Position ($n=1.0$) accurate.	"Not all channels encode quantity equally"

WHEN TO APPLY WHAT — Exam Decision Guide

IF the scenario involves...	APPLY...	KEY NAME
Choosing colour/size/shape for data	Bertin's visual variables → match variable to data type	Bertin
Why a chart is accurate or not	Stevens' law + Cleveland & McGill ranking → state the exponent & rank	Stevens + C&M;
Designing interaction for complex data	Shneiderman's mantra + 7 tasks → overview→filter→details	Shneiderman
Why a diagram helps vs text	Larkin & Simon → informational vs computational equivalence; 3 advantages	Larkin & Simon
User selects/queries data	Heer's generalised selection → declarative query + relaxation vs dynamic queries	Heer
5+ dimensional data	Parallel coordinates or SPLOM → describe brushing + axis reordering	Munzner
Network or graph data	Node-link vs matrix → decide based on graph density + task type	Graph theory
Data changes over time	Line chart / small multiples. State animation vs small multiples argument	Shneiderman/Tufte
Evaluating a vis system	Munzner's nested model (4 levels + threats). Quant vs qual evaluation	Munzner
Choosing a colour map	Sequential / Diverging / Categorical. Never rainbow.	Colour theory
SQL errors / hard training	Dynamic queries → sliders + dropdowns, no syntax errors possible	Shneiderman

**EXAM
STRATEGY**

For every 15-mark answer: (1) Classify the data type → (2) Choose marks & channels, justify with ranking → (3) Apply Shneiderman's mantra to interaction → (4) Name the relevant person/paper → (5) Mention what would go wrong (pitfalls). The exam tests **applied reasoning**, not memorisation. Ask: "How would I **USE** this technique here?"